CBSE Class 12 Physics — Chapterwise Formula Sheet

Here is a complete, exhaustive, and syllabus-aligned (2025-26) formula sheet covering all 14 chapters of the CBSE Class 12 Physics curriculum. Topics recently deleted by CBSE (such as radioactivity, transistors, potentiometers, and logic gates) have been carefully omitted to keep your revision strictly board-focused.

Chapter 1: Electric Charges and Fields

- **Quantization of Charge:** $q = ne$ (where n is an integer, $e = 1.6 \times 10^{-19}\ \mathrm{C}$)
- **Coulomb's Law:**

$$F = \frac{1}{4\pi\varepsilon_0}\frac{|q_1 q_2|}{r^2}$$

  (Electrostatic force between two point charges)
- **Electric Field Intensity:**

$$E = \frac{F}{q_0}$$

  (Force experienced per unit positive test charge)
- **Electric Field due to a Point Charge:**

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$$

- **Electric Dipole Moment:**

$$\vec{p} = q \times 2\vec{a}$$

  (Directed from $-q$ to $+q$)
- **Electric Field on Axial Line of Dipole:**

$$E_{axial} = \frac{1}{4\pi\varepsilon_0}\frac{2pr}{(r^2 - a^2)^2} \approx \frac{1}{4\pi\varepsilon_0}\frac{2p}{r^3}$$

  (For $r \gg a$)
- **Electric Field on Equatorial Line of Dipole:**

$$E_{equatorial} = \frac{1}{4\pi\varepsilon_0}\frac{p}{(r^2 + a^2)^{3/2}} \approx \frac{1}{4\pi\varepsilon_0}\frac{p}{r^3}$$

  (For $r \gg a$)
- **Torque on a Dipole in Uniform E-Field:**

$$\vec{\tau} = \vec{p} \times \vec{E}$$

or $\tau = pE \sin\theta$

- **Electric Flux:**

$$\phi_E = \oint \vec{E} \cdot d\vec{A} = EA \cos\theta$$

- **Gauss's Theorem:**

$$\phi_E = \frac{Q_{enclosed}}{\varepsilon_0}$$

(Total flux through a closed surface)

- **Field due to an Infinitely Long Straight Wire:**

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

(λ = linear charge density)

- **Field due to an Infinite Uniform Plane Sheet:**

$$E = \frac{\sigma}{2\varepsilon_0}$$

(σ = surface charge density)

- **Field due to a Thin Spherical Shell:**
  - Outside ($r > R$):

$$E = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}$$

  - Inside ($r < R$):

$$E = 0$$

Chapter 2: Electrostatic Potential and Capacitance

- **Electric Potential:**

$$V = \frac{W}{q}$$

(Work done per unit charge)

- **Potential due to a Point Charge:**

$$V = \frac{1}{4\pi\varepsilon_0} \frac{q}{r}$$

- **Potential due to an Electric Dipole:**

$$V = \frac{1}{4\pi\varepsilon_0} \frac{p\cos\theta}{r^2 - a^2\cos^2\theta}$$

- **Potential Energy of Two Point Charges:**

$$U = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r}$$

- **Potential Energy of a Dipole in E-Field:**

$$U = -\vec{p} \cdot \vec{E} = -pE\cos\theta$$

- **Relation between E and V:**

$$E = -\frac{dV}{dr}$$

(Electric field is the negative potential gradient)

- **Capacitance:**

$$C = \frac{Q}{V}$$

(SI Unit: Farad, F)

- **Capacitance of a Parallel Plate Capacitor (Air/Vacuum):**

$$C_0 = \frac{\varepsilon_0 A}{d}$$

- **Capacitance with a Dielectric Medium:**

$$C = \frac{K\varepsilon_0 A}{d} = KC_0$$

(K = dielectric constant)

- **Capacitors in Series:**

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

- **Capacitors in Parallel:**

$$C_p = C_1 + C_2 + \dots$$

- **Energy Stored in a Capacitor:**

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$$

- **Energy Density:**

$$u = \frac{1}{2}\varepsilon_0 E^2$$

(Energy stored per unit volume)

Chapter 3: Current Electricity

- **Electric Current:**

$$I = \frac{dq}{dt}$$

(SI Unit: Ampere, A)

- **Drift Velocity:**

$$v_d = \frac{eE}{m}\tau$$

(τ = relaxation time)

- **Relation between Current and Drift Velocity:**

$$I = nAev_d$$

(n = electron number density)

- **Mobility:**

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m}$$

- **Ohm's Law:**

$$V = IR$$

- **Resistance:**

$$R = \rho\frac{l}{A}$$

(ρ = resistivity)

- **Current Density:**

$$J = \frac{I}{A} = \sigma E$$

($\sigma = 1/\rho$ = conductivity)

- **Temperature Dependence of Resistivity:**

$$\rho_T = \rho_0[1 + \alpha(T - T_0)]$$

(α = temp. coefficient)

- **EMF and Terminal Voltage of a Cell:**

$$V = E - Ir$$

(During discharge, r = internal resistance)

- **Internal Resistance formula:**

$$r = \left(\frac{E}{V} - 1\right)R$$

- **Cells in Series:**

$$E_{eq} = E_1 + E_2$$

and

$$r_{eq} = r_1 + r_2$$

- **Cells in Parallel:**

$$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2}$$

and

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2}$$

- **Electrical Power:**

$$P = VI = I^2R = \frac{V^2}{R}$$

- **Wheatstone Bridge (Balanced Condition):**

$$\frac{P}{Q} = \frac{R}{S}$$

(or $\frac{R_1}{R_2} = \frac{R_3}{R_4}$)

Chapter 4: Moving Charges and Magnetism

- **Lorentz Magnetic Force:**

$$\vec{F} = q(\vec{v} \times \vec{B})$$

or

$$F = qvB\sin\theta$$

- **Lorentz Force (Combined E and B fields):**

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

- **Force on a Current-Carrying Conductor:**

$$\vec{F} = I(\vec{l} \times \vec{B})$$

or

$$F = IlB\sin\theta$$

- **Radius of Circular Path in Magnetic Field:**

$$r = \frac{mv}{qB} = \frac{p}{qB}$$

- **Biot-Savart Law:**

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I(d\vec{l} \times \hat{r})}{r^2}$$

- **Magnetic Field at the Center of a Circular Loop:**

$$B = \frac{\mu_0 I}{2R}$$

(For N turns, multiply by N)

- **Magnetic Field on the Axis of a Circular Loop:**

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$

- **Ampere's Circuital Law:**

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enclosed}$$

- **Magnetic Field due to Long Straight Wire:**

$$B = \frac{\mu_0 I}{2\pi r}$$

- **Magnetic Field Inside a Long Solenoid:**

$$B = \mu_0 n I$$

($n = N/L$ is turns per unit length)

- **Force per Unit Length between Parallel Wires:**

$$f = \frac{\mu_0 I_1 I_2}{2\pi d}$$

(Attractive if same direction)

- **Magnetic Dipole Moment of a Current Loop:**

$$\vec{m} = NI\vec{A}$$

- **Torque on a Current Loop:**

$$\vec{\tau} = \vec{m} \times \vec{B}$$

or

$$\tau = NIAB\sin\theta$$

- **Galvanometer to Ammeter (Shunt):**

$$S = \frac{I_g R_g}{I - I_g}$$

(Connected in parallel)

- **Galvanometer to Voltmeter (Series Resistance):**

$$R = \frac{V}{I_g} - R_g$$

(Connected in series)

Chapter 5: Magnetism and Matter

- **Magnetic Dipole Moment of Revolving Electron:**

$$m = \frac{evr}{2}$$

- **Bohr Magneton:**

$$\mu_B = \frac{eh}{4\pi m_e} = 9.27 \times 10^{-24}\ \mathrm{Am}^2$$

(Minimum magnetic moment)

- **Magnetization:**

$$M = \frac{m_{net}}{V}$$

(Magnetic moment per unit volume)

- **Magnetic Intensity:**

$$H = \frac{B}{\mu_0} - M$$

- **Magnetic Susceptibility:**

$$M = \chi H$$

- **Relative Permeability:**

$$\mu_r = 1 + \chi$$

and

$$B = \mu_0 \mu_r H$$

Chapter 6: Electromagnetic Induction

- **Magnetic Flux:**

$$\phi_B = \vec{B} \cdot \vec{A} = BA\cos\theta$$

- **Faraday's Law of Induction:**

$$\varepsilon = -N\frac{d\phi_B}{dt}$$

(Induced EMF)

- **Motional EMF (Straight Conductor):**

$$\varepsilon = Blv$$

- **Motional EMF (Rotating Rod):**

$$\varepsilon = \frac{1}{2}B\omega l^2$$

- **Self-Inductance:**

$$\phi = LI$$

and

$$\varepsilon = -L\frac{dI}{dt}$$

- **Self-Inductance of a Long Solenoid:**

$$L = \frac{\mu_0 N^2 A}{l}$$

- **Energy Stored in an Inductor:**

$$U = \frac{1}{2} L I^2$$

- **Mutual Inductance:**

$$\phi_2 = M I_1$$

and

$$\varepsilon_2 = -M \frac{dI_1}{dt}$$

- **Mutual Inductance of Coaxial Solenoids:**

$$M = \frac{\mu_0 N_1 N_2 A}{l}$$

Chapter 7: Alternating Current

- **Instantaneous Voltage and Current:**

$$V = V_0 \sin(\omega t)$$

and

$$I = I_0 \sin(\omega t \pm \phi)$$

- **RMS Values:**

$$V_{rms} = \frac{V_0}{\sqrt{2}}$$

and

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

- **Inductive Reactance:**

$$X_L = \omega L = 2\pi f L$$

(Current lags voltage by 90°)

- **Capacitive Reactance:**

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

(Current leads voltage by 90°)

- **Impedance of Series LCR Circuit:**

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

- **Phase Angle in LCR:**

$$\tan\phi = \frac{X_L - X_C}{R}$$

- **Resonant Frequency:**

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

(Occurs when $X_L = X_C$)

- **Average Power:**

$$P_{avg} = V_{rms} I_{rms} \cos\phi$$

- **Power Factor:**

$$\cos\phi = \frac{R}{Z}$$

- **Transformer Turns Ratio:**

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

- **Transformer Efficiency:**

$$\eta = \frac{V_s I_s}{V_p I_p} \times 100\%$$

Chapter 8: Electromagnetic Waves

- **Displacement Current:**

$$I_d = \varepsilon_0 \frac{d\phi_E}{dt}$$

- **Ampere-Maxwell Law:**

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d)$$

- **Speed of EM Wave in Vacuum:**

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 3 \times 10^8 \text{ m/s}$$

- **Speed of EM Wave in a Medium:**

$$v = \frac{1}{\sqrt{\mu \varepsilon}}$$

- **Relation between E and B amplitudes:**

$$c = \frac{E_0}{B_0}$$

- **Momentum of EM Wave (Photon):**

$$p = \frac{U}{c}$$

(Where U is energy)

Chapter 9: Ray Optics and Optical Instruments

- **Mirror Formula:**

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

- **Linear Magnification (Mirror):**

$$m = -\frac{v}{u} = \frac{h_i}{h_o}$$

- **Snell's Law of Refraction:**

$$n_1 \sin i = n_2 \sin r$$

or

$$\frac{\sin i}{\sin r} =^1 n_2 = \frac{n_2}{n_1} = \frac{v_1}{v_2}$$

- **Critical Angle (Total Internal Reflection):**

$$\sin i_c = \frac{n_R}{n_D} = \frac{1}{^R n_D}$$

- **Refraction at Spherical Surface:**

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

- **Lens Maker's Formula:**

$$\frac{1}{f} = (n_{21} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

- **Thin Lens Formula:**

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

- **Linear Magnification (Lens):**

$$m = \frac{v}{u} = \frac{h_i}{h_o}$$

- **Power of a Lens:**

$$P = \frac{1}{f(\text{in meters})}$$

  (SI Unit: Diopter, D)

- **Combination of Thin Lenses:**

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \dots$$

  and

$$P = P_1 + P_2 + \dots$$

- **Prism Formula:**

$$n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

  (A = angle of prism, δ_m = minimum deviation)

- **Magnifying Power of Simple Microscope:**

$$M = 1 + \frac{D}{f}$$

  (Image at near point $D = 25\ \text{cm}$)

- **Magnifying Power of Compound Microscope:**

$$M \approx \frac{L}{f_o}\left(1 + \frac{D}{f_e}\right)$$

(Image at near point, L = tube length)

- **Astronomical Telescope (Normal Adjustment):**

$$M = \frac{f_o}{f_e}$$

and Tube Length

$$L = f_o + f_e$$

Chapter 10: Wave Optics

- **Relation between Phase Difference and Path Difference:**

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$$

- **Resultant Intensity of two Coherent Waves:**

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

- **Constructive Interference (Maxima):** Path difference

$$\Delta x = n\lambda$$

($n = 0, 1, 2 . . .$)

- **Destructive Interference (Minima):** Path difference

$$\Delta x = (n - 0.5)\lambda$$

($n = 1, 2 . . .$)

- **Fringe Width in YDSE:**

$$\beta = \frac{\lambda D}{d}$$

- **Single Slit Diffraction (Width of Central Maximum):**

$$W = \frac{2\lambda D}{a}$$

- **Single Slit Diffraction (Minima Condition):**

$$a\sin\theta = n\lambda$$

(a = slit width)

Chapter 11: Dual Nature of Radiation and Matter

- **Energy of a Photon:**

$$E = h\nu = \frac{hc}{\lambda}$$

(Planck's Constant $h = 6.63 \times 10^{-34}$ Js)

- **Momentum of a Photon:**

$$p = \frac{h}{\lambda} = \frac{E}{c}$$

- **Work Function:**

$$\phi_0 = h\nu_0 = \frac{hc}{\lambda_0}$$

(ν_0 = threshold frequency)

- **Einstein's Photoelectric Equation:**

$$K_{max} = h\nu - \phi_0 = h(\nu - \nu_0)$$

- **Stopping Potential:**

$$eV_0 = K_{max}$$

$\implies$

$$V_0 = \frac{h\nu}{e} - \frac{\phi_0}{e}$$

- **de-Broglie Wavelength:**

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

- **de-Broglie Wavelength in terms of Kinetic Energy:**

$$\lambda = \frac{h}{\sqrt{2mK}}$$

- **de-Broglie Wavelength of an Accelerated Electron:**

$$\lambda = \frac{h}{\sqrt{2meV}} \approx \frac{12.27}{\sqrt{V}} \backslash \mathrm{AA}$$

Chapter 12: Atoms

- **Distance of Closest Approach:**

$$r_0 = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K}$$

(K = initial kinetic energy of α-particle)

- **Bohr's Quantization Postulate:**

$$mvr = \frac{nh}{2\pi}$$

(Angular momentum is quantized)

- **Radius of n^{th} Orbit (Hydrogen):**

$$r_n = \frac{\varepsilon_0 h^2 n^2}{\pi m e^2} = 0.529 n^2 \ \backslash\mathrm{AA}$$

- **Velocity of Electron in n^{th} Orbit:**

$$v_n = \frac{e^2}{2\varepsilon_0 h n} = \frac{c}{137n}$$

- **Total Energy in n^{th} Orbit:**

$$E_n = -\frac{me^4}{8\varepsilon_0^2 h^2 n^2} = -\frac{13.6}{n^2}\ \mathrm{eV}$$

- **Kinetic and Potential Energy Relations:**

$$K = -E_n$$

and

$$U = 2E_n$$

- **Rydberg Formula (Spectral Lines):**

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

($R_H \approx 1.097 \times 10^7\ \mathrm{m}^{-1}$)

Chapter 13: Nuclei

- **Nuclear Radius:**

$$R = R_0 A^{1/3}$$

(where $R_0 = 1.2 \times 10^{-15}\ \mathrm{m} = 1.2\ \mathrm{fm}$, A = Mass Number)

- **Nuclear Density:**

$$\rho \approx 2.3 \times 10^{17}\ \mathrm{kg/m^3}$$

(Independent of mass number A)

- **Mass-Energy Equivalence:**

$$E = \Delta mc^2$$

- **Mass Defect:**

$$\Delta m = [Zm_p + (A - Z)m_n] - M_{nucleus}$$

- **Nuclear Binding Energy:**

$$BE = \Delta m \times c^2 = \Delta m \times 931.5\ \mathrm{MeV/u}$$

Chapter 14: Semiconductor Electronics

- **Intrinsic Carrier Concentration:**

$$n_i^2 = n_e \times n_h$$

(Mass-action law)

- **Total Current in a Semiconductor:**

$$I = I_e + I_h$$

(Sum of electron and hole currents)

- **Conductivity of a Semiconductor:**

$$\sigma = e(n_e\mu_e + n_h\mu_h)$$

- **Dynamic Resistance of a Junction Diode:**

$$r_d = \frac{\Delta V}{\Delta I}$$

(Ratio of small change in voltage to small change in current)